Unit 5 Test Study Guide DRAFT

Systems of Equations and Inequalities | Draft New Format

Name: ______________________________ **Period:** __________ **Due:** ____________

How to use this guide: This draft follows the Unit 5 test structure: solution checks, number of solutions, solving systems by graphing/substitution/elimination, systems from contexts, and systems of inequalities. Each skill has a Version A and Version B so students can try a second similar problem after getting help.

Multiple-Choice Practice

1A. Checking Solutions | Version A

Is (2, 5) a solution to $y = 2x + 1$ and $x + y = 7$? A. Yes B. No C. Only first equation D. Only second equation	Work / thinking __________________________ __________________________ __________________________

1B. Checking Solutions | Version B

Is (-1, 4) a solution to $y = -3x + 1$ and $2x + y = 2$? A. Yes B. No C. Only first equation D. Only second equation	Work / thinking __________________________ __________________________ __________________________

2A. Solving by Inspection | Version A

What is the solution of $y = x + 3$ and $y = -x + 7$? A. (2, 5) B. (5, 2) C. (0, 3) D. no solution	Work / thinking __________________________ __________________________ __________________________

2B. Solving by Inspection | Version B

What is the solution of $y = 2x - 1$ and $y = -x + 8$? A. (2, 5) B. (3, 5) C. (5, 3) D. no solution	Work / thinking __________________________ __________________________ __________________________

3A. Number of Solutions | Version A

How many solutions does $y = 3x + 2$ and $y = 3x - 5$ have? A. one B. two C. none D. infinitely many	Work / thinking ________________________ ________________________ ________________________

3B. Number of Solutions | Version B

How many solutions does $y = -2x + 4$ and $2x + y = 4$ have? A. one B. none C. infinitely many D. two	Work / thinking ________________________ ________________________ ________________________

4A. Substitution Step | Version A

For $y = 2x - 3$ and $x + y = 12$, which substitution equation is valid? A. $x + 2x - 3 = 12$ B. $2x - 3 = 12$ C. $x + y = 2x - 3$ D. $2x + 12 = -3$	Work / thinking ________________________ ________________________ ________________________

4B. Substitution Step | Version B

For $x = y + 4$ and $3x - y = 10$, which substitution equation is valid? A. $3(y + 4) - y = 10$ B. $y + 4 = 10$ C. $3x - (y + 4) = 10$ D. $x = 3y - 10$	Work / thinking ________________________ ________________________ ________________________

5A. Elimination | Version A

Which variable is eliminated if you add 4x + 3y = 20 and -4x + y = -8? A. x B. y C. both D. neither	Work / thinking ______________________ ______________________ ______________________

5B. Elimination | Version B

Which variable is eliminated if you add 2x - 5y = 1 and 3x + 5y = 14? A. x B. y C. both D. neither	Work / thinking ______________________ ______________________ ______________________

6A. Systems from Context | Version A

Adult tickets cost $8, student tickets cost $5, and 40 tickets cost $260. Which system fits? A. a + s = 40; 8a + 5s = 260 B. a + s = 260; 8a + 5s = 40 C. 8a + s = 260; a + 5s = 40 D. a - s = 40; 8a - 5s = 260	Work / thinking ______________________ ______________________ ______________________

6B. Systems from Context | Version B

A shelter has cats and dogs. Cats are 4 more than twice dogs, and there are 34 animals. Which system fits? A. c = 2d + 4; c + d = 34 B. d = 2c + 4; c + d = 34 C. c = 4d + 2; c + d = 34 D. c + d = 4; c = 34	Work / thinking ______________________ ______________________ ______________________

7A. Testing Inequality Systems | Version A

Does (3, 2) satisfy $y \le x$ and $y > -x + 1$? A. Yes B. No C. first only D. second only	Work / thinking ______________________________ ______________________________ ______________________________

7B. Testing Inequality Systems | Version B

Does (0, -1) satisfy $y \ge 2x - 3$ and $y < x + 2$? A. Yes B. No C. first only D. second only	Work / thinking ______________________________ ______________________________ ______________________________

8A. Solution Region | Version A

For $y > x + 1$ and $y \le -2x + 7$, which point is in the solution set? A. (0, 0) B. (1, 4) C. (3, 0) D. (4, 6)	Work / thinking ______________________________ ______________________________ ______________________________

8B. Solution Region | Version B

For $y \ge -x + 2$ and $y < 3x + 1$, which point is in the solution set? A. (0, 0) B. (1, 1) C. (2, 3) D. (-1, 5)	Work / thinking ______________________________ ______________________________ ______________________________

9A. Classifying Systems | Version A

A system has slopes 2 and -1. How many solutions does it have? A. no solution B. exactly one C. infinitely many D. cannot tell	Work / thinking ________________________ ________________________ ________________________

9B. Classifying Systems | Version B

A system has the same slope and same y-intercept. How many solutions does it have? A. no solution B. exactly one C. infinitely many D. two	Work / thinking ________________________ ________________________ ________________________

10A. Interpreting Solutions | Version A

If a system solution is (6, 14) where x is notebooks and y is total cost, what does 14 represent? A. 14 notebooks B. $14 total cost C. 14 dollars per notebook D. 14 equations	Work / thinking ________________________ ________________________ ________________________

10B. Interpreting Solutions | Version B

If a system solution is (5, 11) where x is hours and y is money earned, what does 5 represent? A. 5 hours B. $5 C. $5 per hour D. 5 workers	Work / thinking ________________________ ________________________ ________________________

Free-Response Practice

Show clear work. Use the Version B problem as a second check after you have tried Version A.

11-A. Graphing Systems | Version A

Solve by graphing or substitution: $y = x + 2$ and $y = -x + 8$. **Answer:** ____________________	Work / thinking ____________________________ ____________________________ ____________________________ ____________________________ ____________________________ ____________________________

11-B. Graphing Systems | Version B

Solve by graphing or substitution: $y = 2x - 1$ and $y = -x + 11$. **Answer:** ____________________	Work / thinking ____________________________ ____________________________ ____________________________ ____________________________ ____________________________ ____________________________

12-A. Substitution | Version A

Solve using substitution: $y = 3x - 4$ and $x + y = 12$. **Answer:** ____________________	Work / thinking ____________________________ ____________________________ ____________________________ ____________________________ ____________________________ ____________________________

12-B. Substitution | Version B

Solve using substitution: $x = y - 1$ and $2x + y = 14$. **Answer:** ____________________	Work / thinking ____________________________ ____________________________ ____________________________ ____________________________ ____________________________ ____________________________

13-A. Elimination | Version A

Solve using elimination: $2x + 3y = 19$ and $2x - y = 7$. **Answer:** ____________________	Work / thinking ___________________________ ___________________________ ___________________________ ___________________________ ___________________________ ___________________________

13-B. Elimination | Version B

Solve using elimination: $3x + 2y = 24$ and $x - 2y = -4$. **Answer:** ____________________	Work / thinking ___________________________ ___________________________ ___________________________ ___________________________ ___________________________ ___________________________

14-A. Systems of Inequalities | Version A

Describe the graph of $y \geq x - 2$ and $y < -2x + 7$. **Answer:** ____________________	Work / thinking ___________________________ ___________________________ ___________________________ ___________________________ ___________________________ ___________________________

14-B. Systems of Inequalities | Version B

Describe the graph of $y > -x + 3$ and $y \leq 2x - 1$. **Answer:** ____________________	Work / thinking ___________________________ ___________________________ ___________________________ ___________________________ ___________________________ ___________________________

15-A. Context System | Version A

Coffee costs \$1.50 and doughnuts cost \$0.75. On Monday, 10 coffees and 5 doughnuts cost \$18.75. On Tuesday, 6 coffees and 12 doughnuts cost \$18.00. Write and solve for the price of each item. **Answer:** ____________________	Work / thinking ___________________________ ___________________________ ___________________________ ___________________________ ___________________________ ___________________________

15-B. Context System | Version B

Adult tickets and student tickets were sold. 8 adult and 6 student tickets cost \$86. 5 adult and 10 student tickets cost \$85. Find the price of each ticket. **Answer:** ____________________	Work / thinking ___________________________ ___________________________ ___________________________ ___________________________ ___________________________ ___________________________

16-A. Coin Problem | Version A

There are 50 nickels and dimes worth \$4.15. How many of each coin are there? **Answer:** ____________________	Work / thinking ___________________________ ___________________________ ___________________________ ___________________________ ___________________________ ___________________________

16-B. Coin Problem | Version B

There are 40 nickels and quarters worth \$6.20. How many of each coin are there? **Answer:** ____________________	Work / thinking ___________________________ ___________________________ ___________________________ ___________________________ ___________________________ ___________________________

Student Self-Check Answer Bank

Important: This is not a worked solution key. It is a self-check bank. If your answer is not here, go back and check the setup, notation, and calculation.

Multiple-Choice Practice

Solve first, then check. Cross off each answer only after your work supports it.

A. $3(y + 4) - y = 10$	B. y
C. infinitely many	A. x
B. exactly one	C. (2, 3)
A. Yes	C. infinitely many
B. $14 total cost	A. $a + s = 40$; $8a + 5s = 260$
C. none	B. (1, 4)
A. Yes	B. (3, 5)
A. 5 hours	A. $x + 2x - 3 = 12$
A. (2, 5)	A. Yes
A. Yes	A. $c = 2d + 4$; $c + d = 34$

Free-Response Practice

Solve first, then check. Cross off each answer only after your work supports it.

Solid $y = x - 2$ shaded above; dashed $y = -2x + 7$ shaded below; overlap is solution	(4, 8)
(3, 5)	Dashed $y = -x + 3$ shaded above; solid $y = 2x - 1$ shaded below; overlap is solution
(4, 5)	19 nickels and 21 quarters
(5, 4.5)	Coffee: $1.50; doughnut: $0.75
(5, 3)	Adult: $7; student: $5
17 nickels and 33 dimes	(4, 7)

Final Check Before the Assessment

- ☐ I can check whether an ordered pair solves a system.
- ☐ I can identify one solution, no solution, and infinitely many solutions.
- ☐ I can solve systems by graphing, substitution, and elimination.
- ☐ I can write systems from contexts and interpret the solution.
- ☐ I can graph and test systems of linear inequalities.